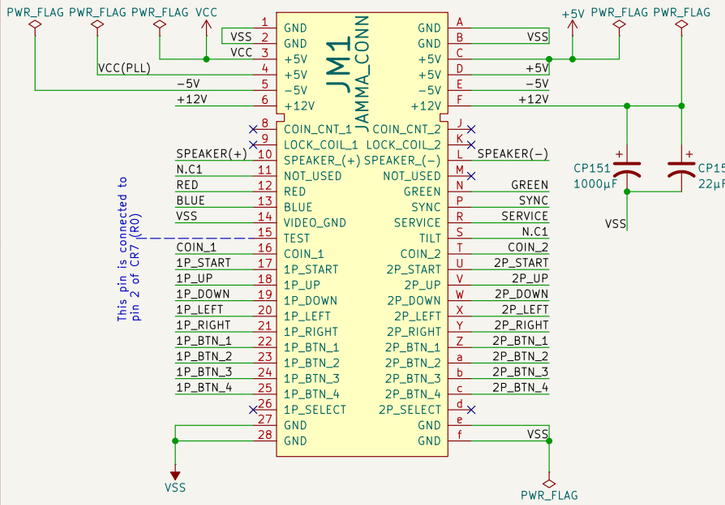
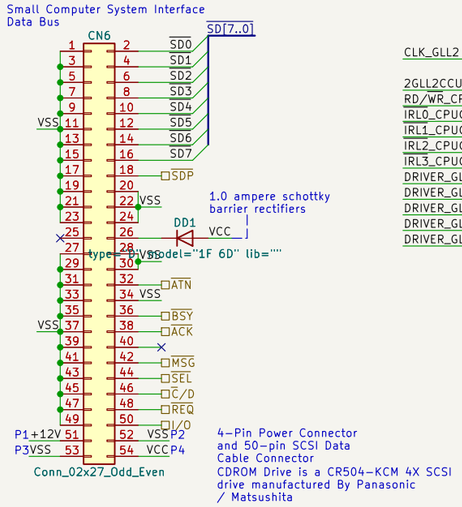
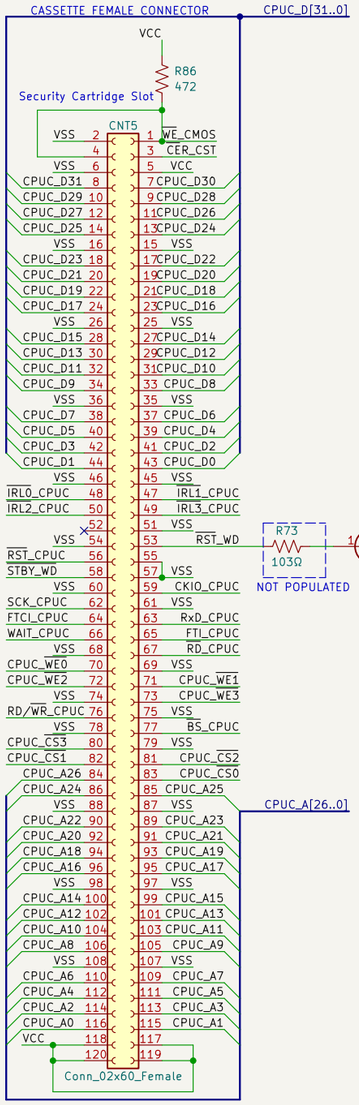
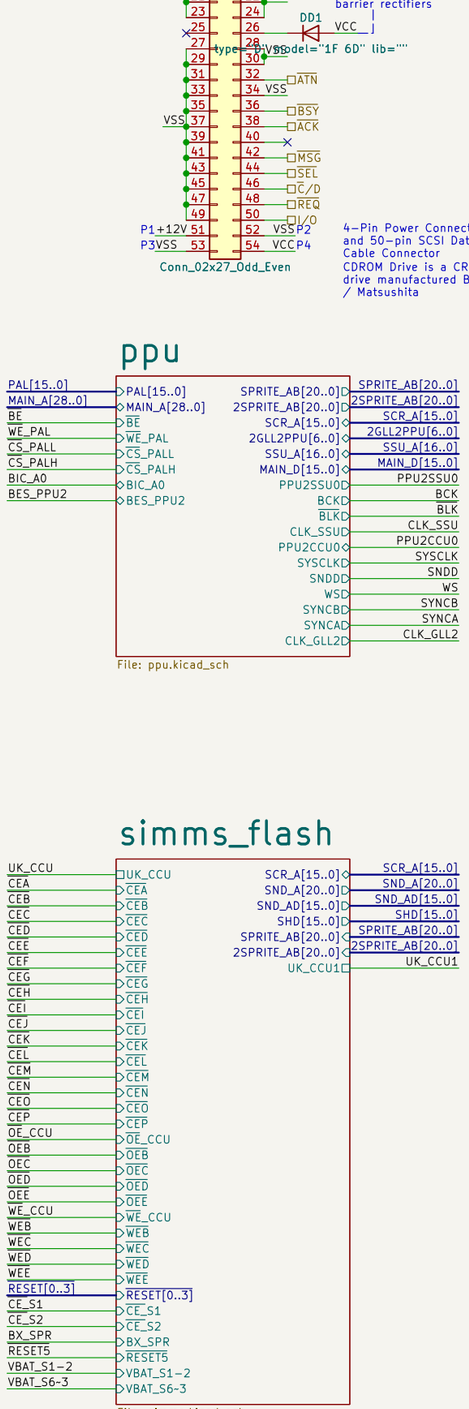
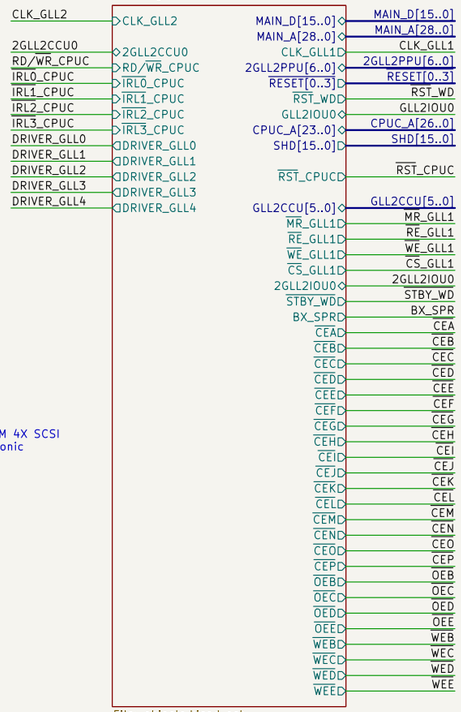
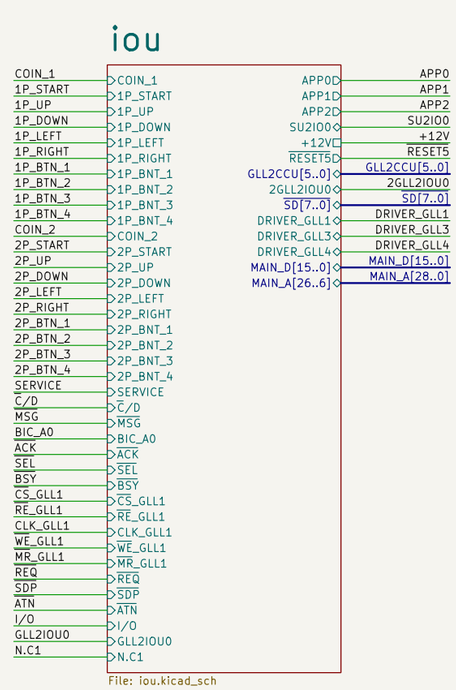




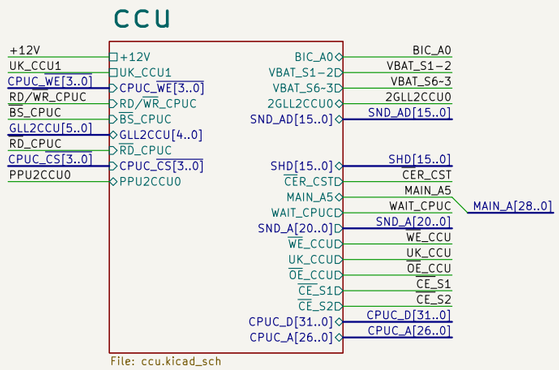




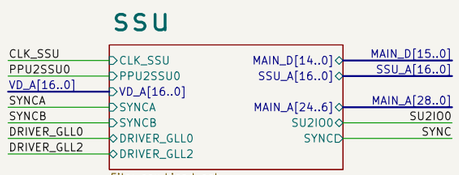


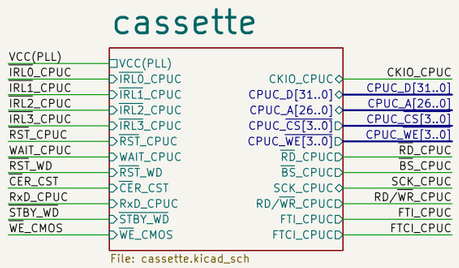


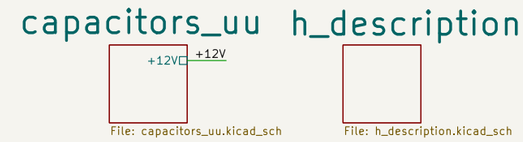








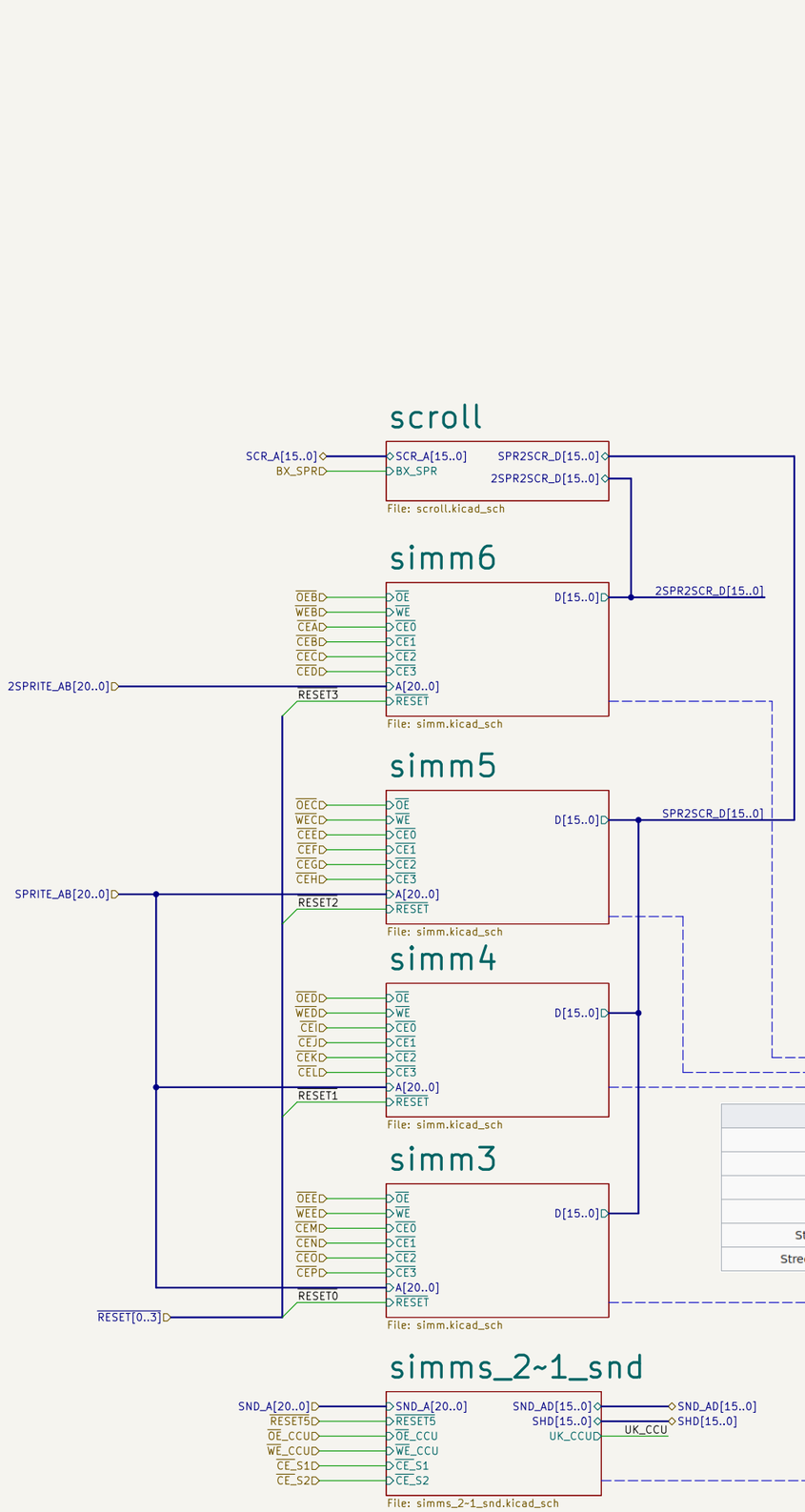




14/10/22−30/01/23
JOTEGO
JOTEGO
Sheet: /
File: cps3.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 1/23



Juego	Ranura 1	Ranura 2	Ranura 3	Ranura 4	Ranura 5	Ranura 6
Warzard / Red Earth	64MBit		128MBit	128MBit	32MBit o 128MBit	
Street Fighter III: New Generation	64MBit		128MBit	128MBit	32MBit o 128MBit	
JoJo's Venture	64MBit	64MBit	128MBit	128MBit	32MBit o 128MBit	
JoJo's Bizarre Adventure	64MBit	64MBit	128MBit	128MBit	128MBit	
Street Fighter III: 2nd Impact Giant Attack	64MBit	64MBit	128MBit	128MBit	128MBit	
Street Fighter III: 3rd Strike Fight for the Future	64MBit	64MBit	128MBit	128MBit	128MBit	128MBit

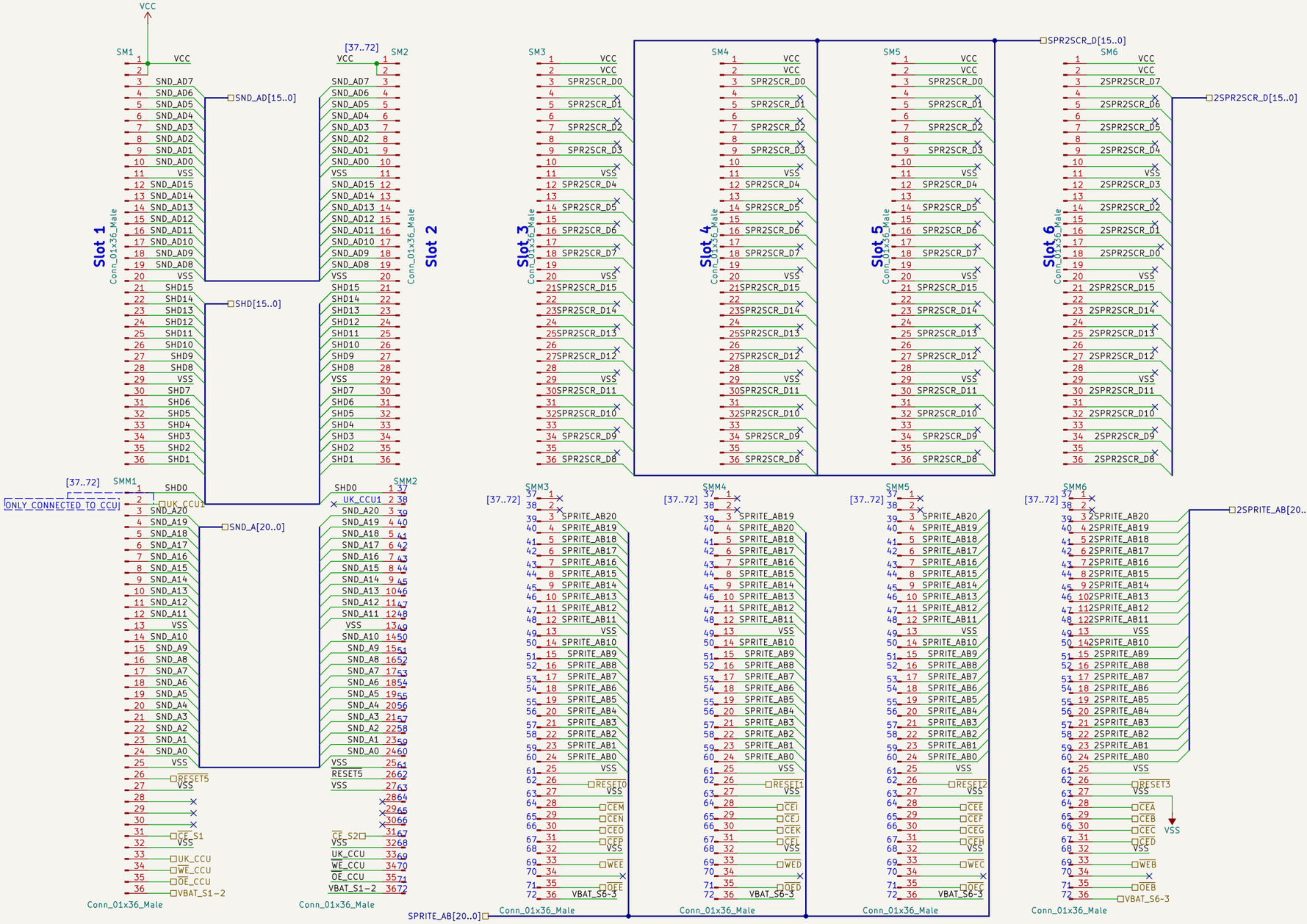
14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /simms_flash/
File: simms.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 2/23

SIMM CONNECTOR

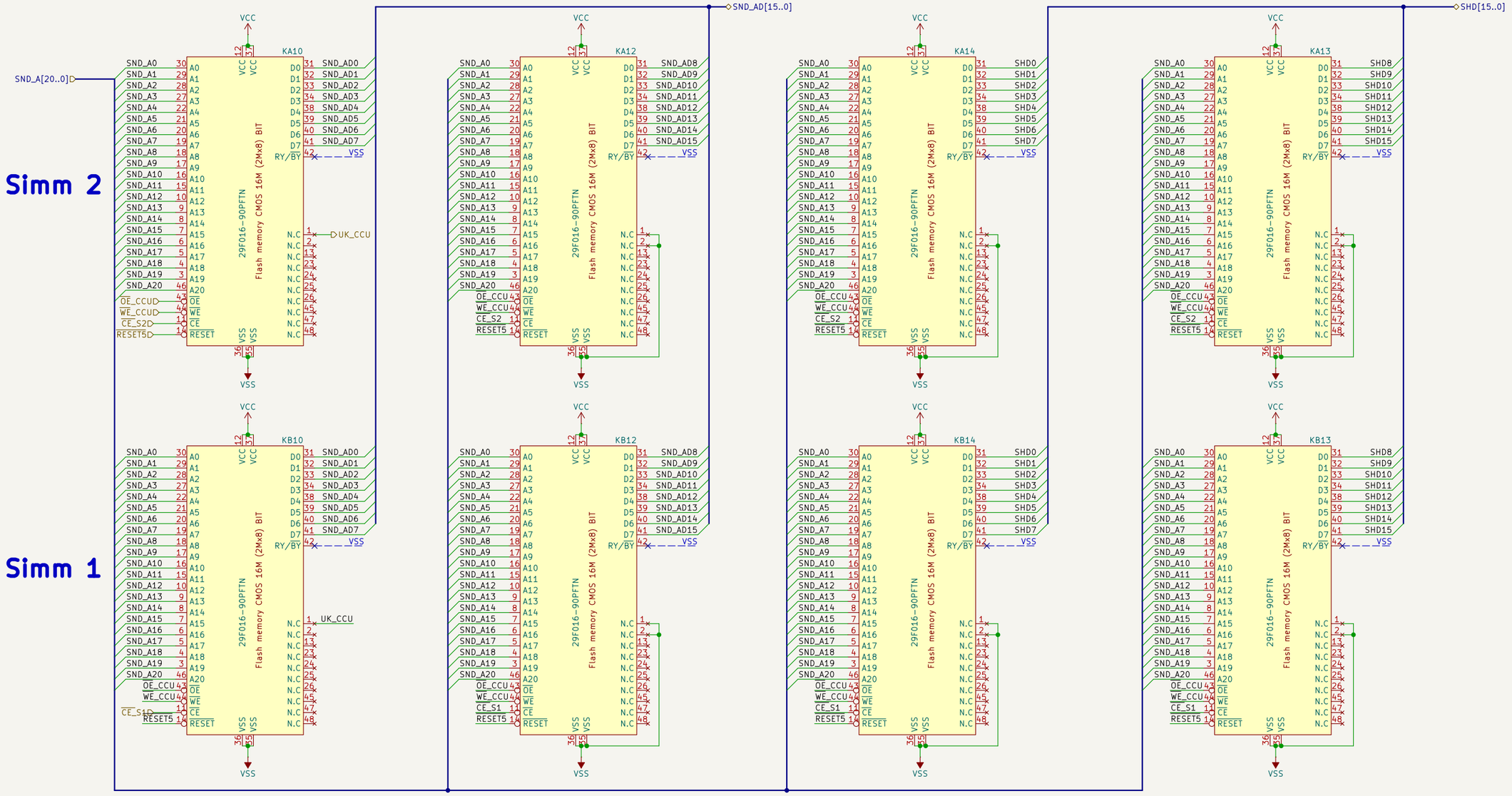


14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /simms_flash/simm_connector/
File: simm_connector.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 12/23



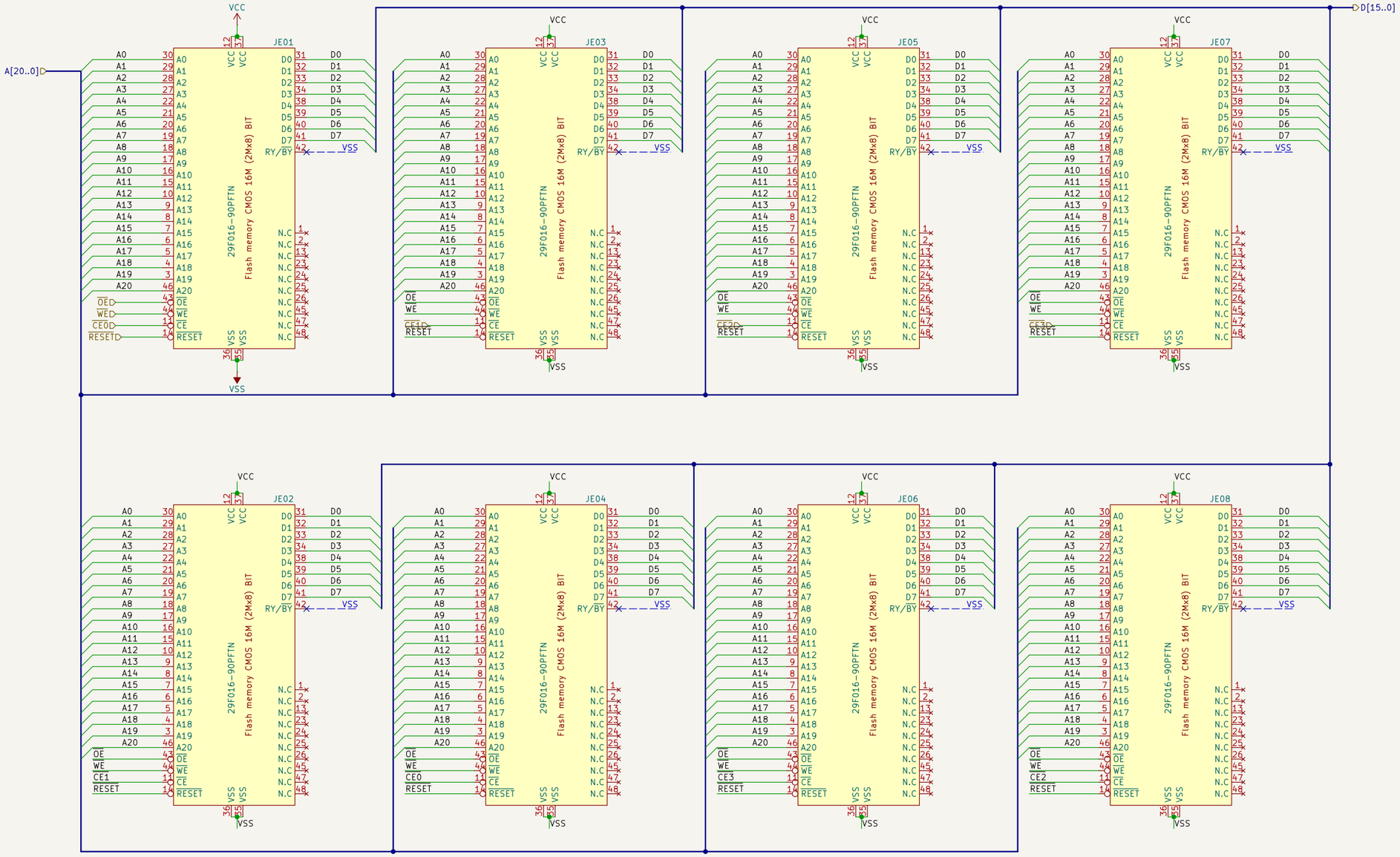
14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /simms_flash/simms_2-1_snd/
File: simms_2-1_snd.kicad_sch

Title: CP SYSTEM III

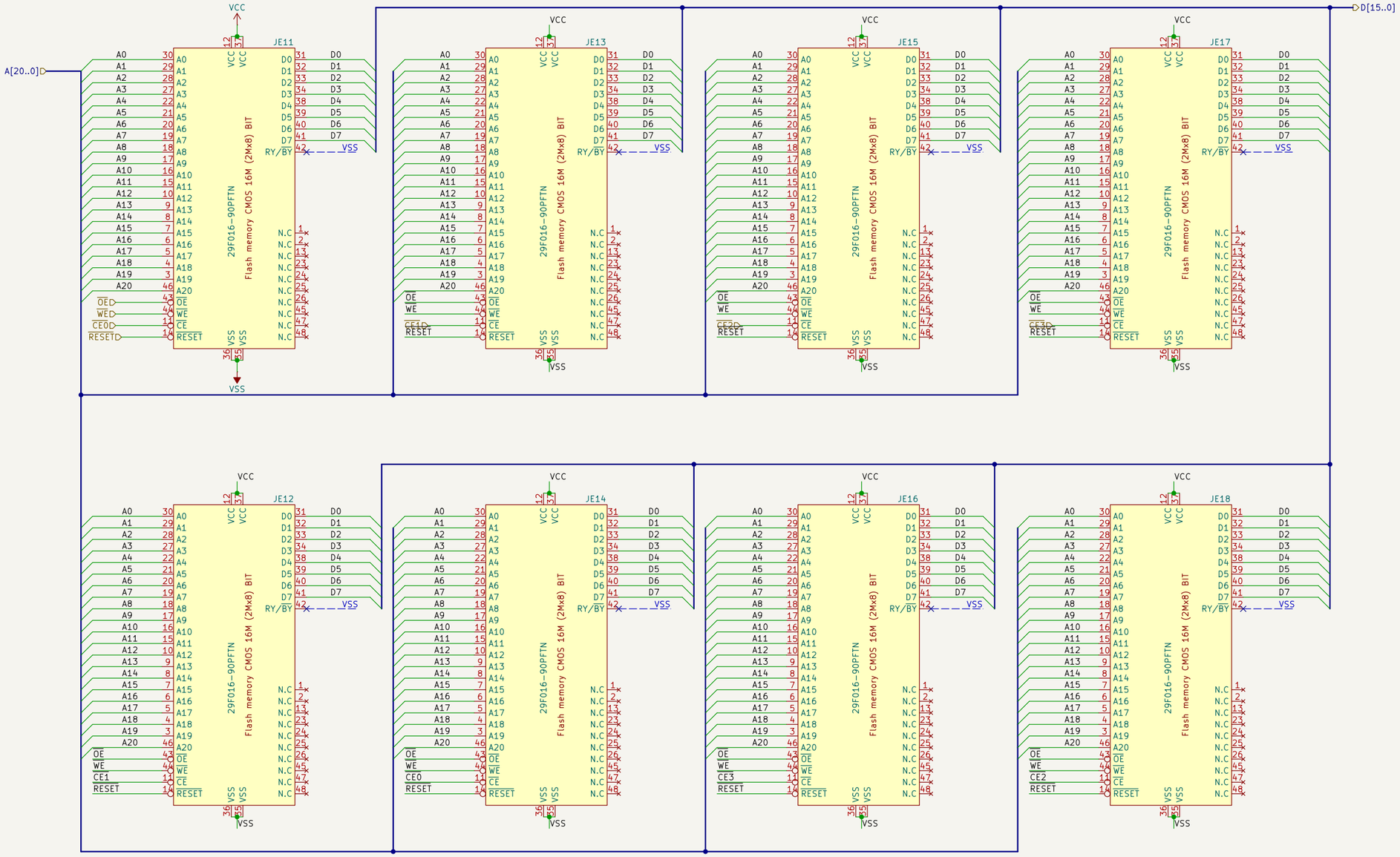
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 13/23

Simm



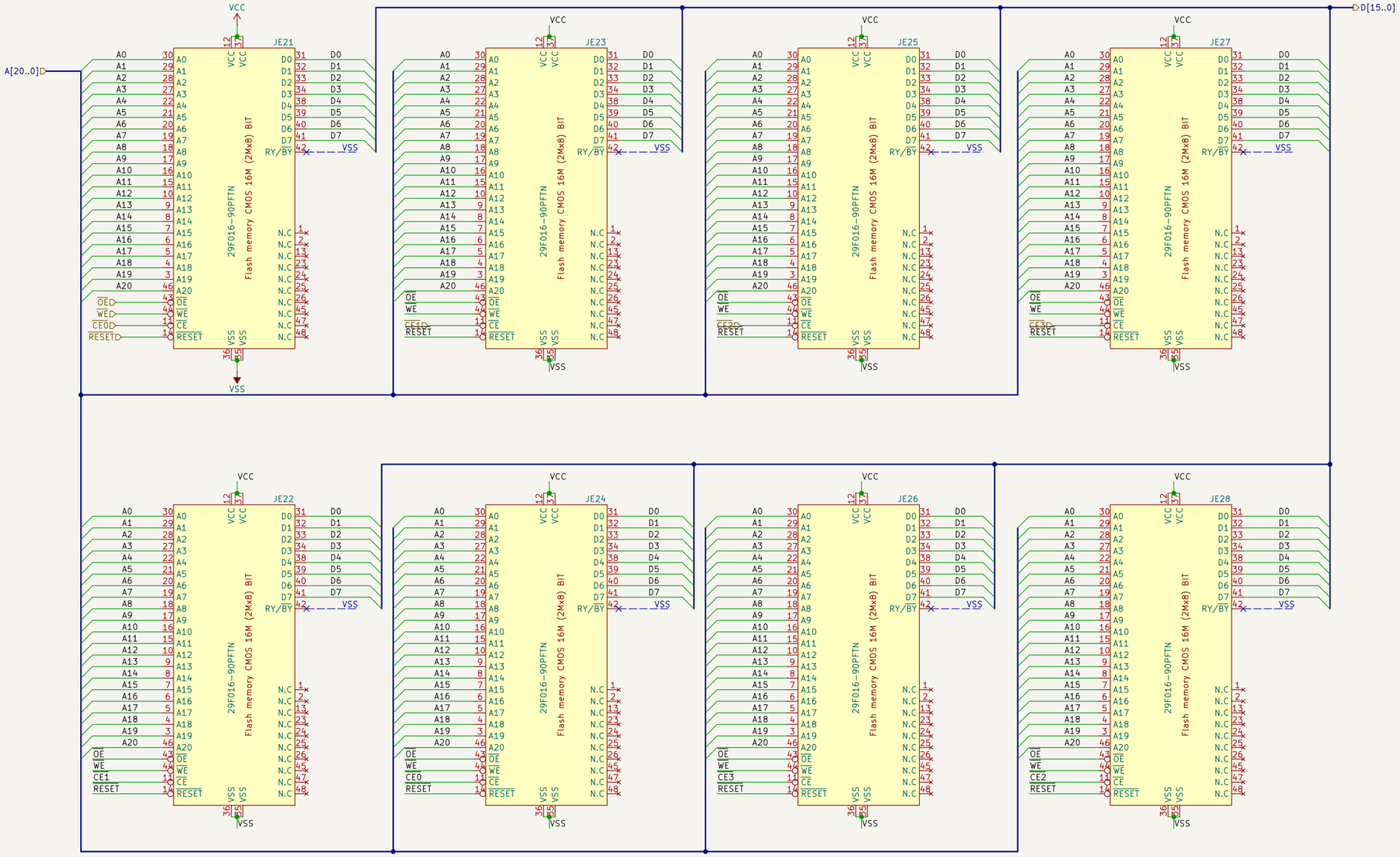
14/10/22–30/01/23		
JOTEGO		
JOTEGO		
Sheet: /simms_flash/simm3/		
File: simm.kicad_sch		
Title: CP SYSTEM III		
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 14/23

Simm



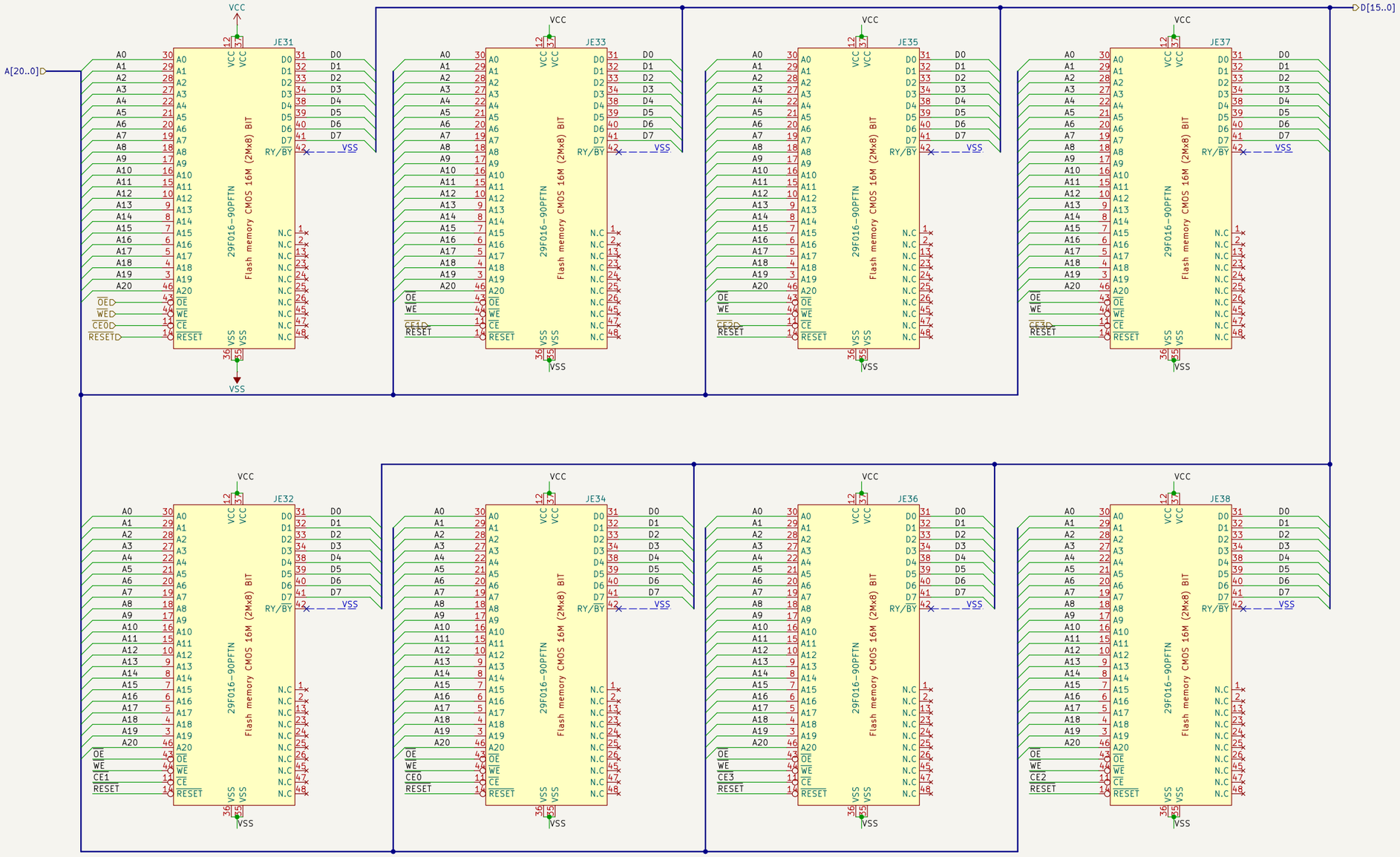
14/10/22–30/01/23		
JOTEGO		
JOTEGO		
Sheet: /simms_flash/simm4/		
File: simm.kicad_sch		
Title: CP SYSTEM III		
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 15/23

Simm



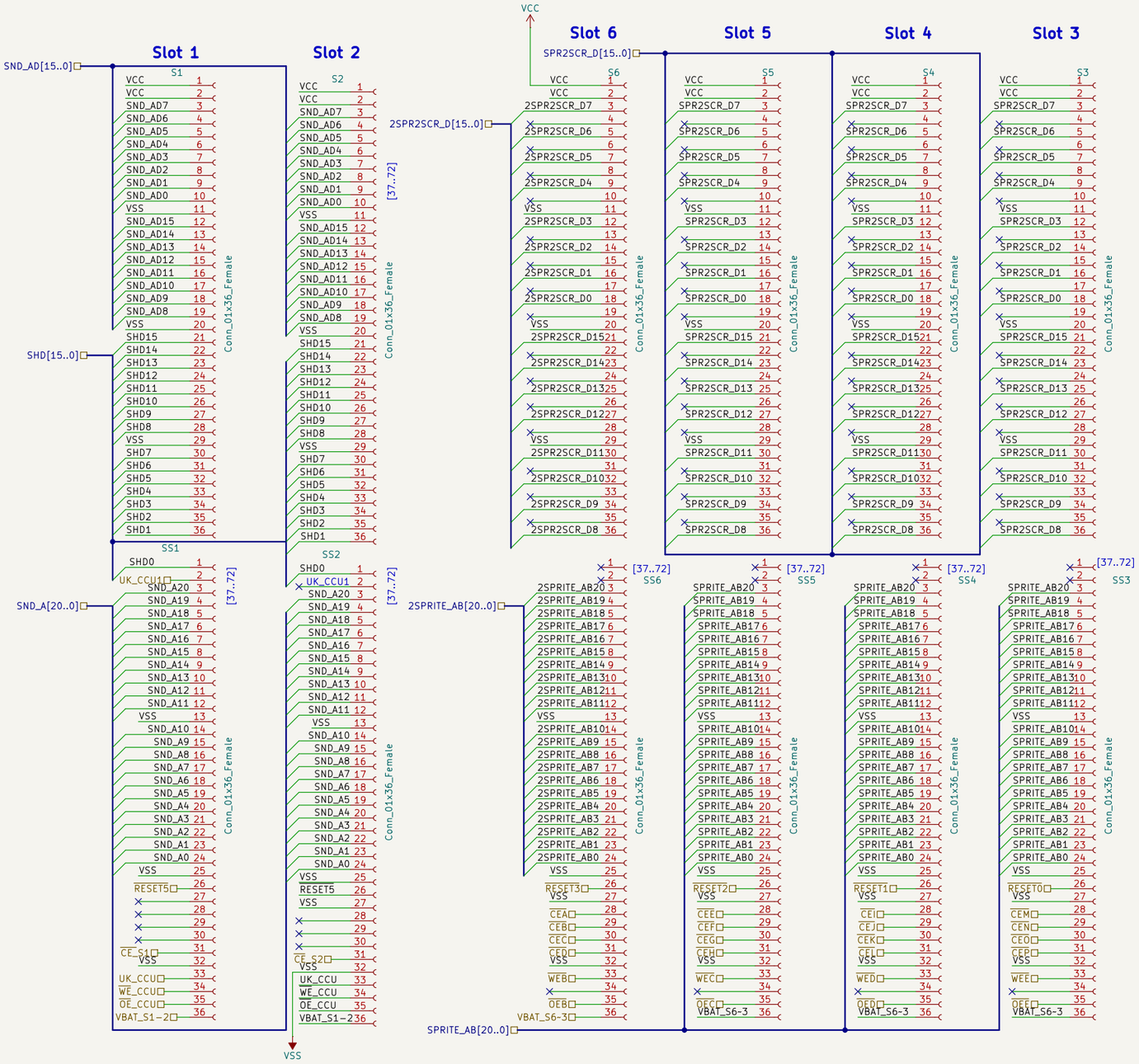
14/10/22–30/01/23		
JOTEGO		
JOTEGO		
Sheet: /simms_flash/simm5/		
File: simm.kicad_sch		
Title: CP SYSTEM III		
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 16/23

Simm



14/10/22–30/01/23		
JOTEGO		
JOTEGO		
Sheet: /simms_flash/simm6/		
File: simm.kicad_sch		
Title: CP SYSTEM III		
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 17/23

Slot Female

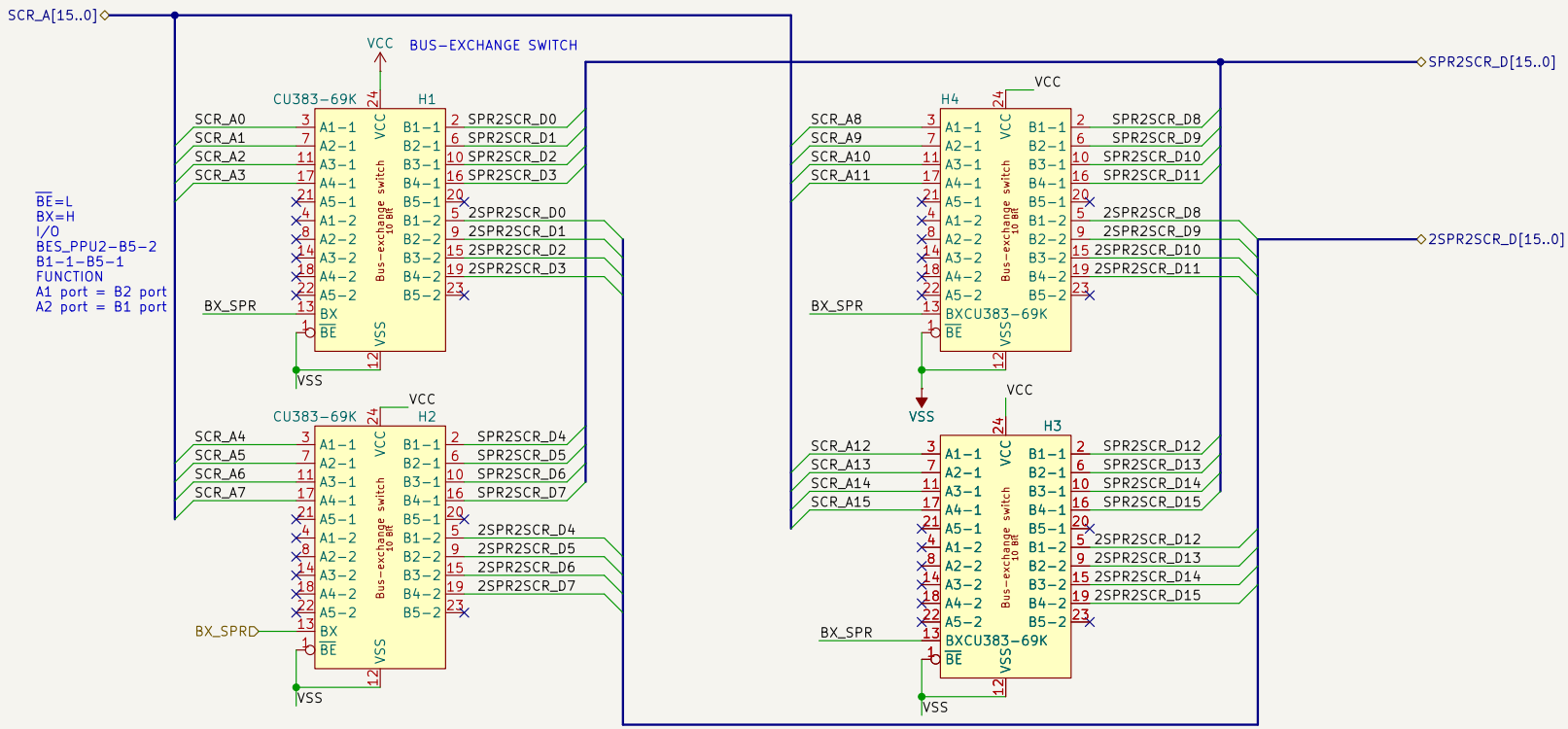


14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /simms_flash/slots_female/
File: slots_females.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 18/23



14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /simms_flash/scroll/
File: scroll.kicad_sch

Title: CP SYSTEM III

Size: A4	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 19/23

F11
DL-2929-IOU
INPUT OUTPUT UNIT
I/O controller.

BUS INTERFACE CONTROLLER

BA
R77 330
R76 105
XTAL3 3.6864MHz
CP CP0
CP CP1
VSS
GLL2IOU0
R84 101

CRE438-60CR9
K_H24
K_H22
K_H20
K_H18

CRE438-60CR8
K_H16
K_H14
2P_BTN5
KH_2P_BTN4

CRE438-60 CR10
KH_1P_BTN4
1P_BTN5
1P_BTN6
K_H13

CRE438-60CR1
2P_BNT_4
2P_BNT_3
2P_BNT_2
2P_BNT_1

CRE438-60CR2
2P_RIGHT
2P_LEFT
2P_DOWN
2P_UP

CRE438-60CR3
1P_BNT_4
1P_BNT_3
1P_BNT_2
1P_BNT_1

CRE438-60CR4
1P_RIGHT
1P_LEFT
1P_DOWN
1P_UP

CRE438-60CR5
K_H12
K_H11
1P_START
2P_START

CRE438-60CR6
K_H10
2P_BTN6
COIN_2
COIN_1

CRE438-60 CR11
K_H30
K_H29

This pin is connected to pin 15 of the jamma (N.C)
CRE438-60 CR7
N.C1
SERVICE

SW1
GP1
SW_MEC_5G
TEST
VSS

VCC
GLL2IOU0
XTAL_3.6864MHz
UK
K_H24
K_H22
K_H20
K_H18
K_H16
K_H14
2P_BTN5
KH_2P_BTN4
KH_1P_BTN4
1P_BTN5
1P_BTN6
K_H13
2P_BTN4
2P_BTN3
2P_BTN2
2P_BTN1
2P_RIGHT
2P_LEFT
2P_DOWN
2P_UP
1P_BTN4
1P_BTN3
1P_BTN2
1P_BTN1
1P_RIGHT
1P_LEFT
1P_DOWN
1P_UP
K_H12
K_H11
1P_STAR
2P_STAR
K_H10
2P_BTN6
COIN_2
COIN_1
K_H29
K_H30
N.C_JAMMA
N.C1_JAMMA
SERVICE
DRIVER_GLL1
DRIVER_GLL3
DRIVER_GLL4
VSS

Application mode 0 APP0
DAC SND Application mode 1 APP1
Application mode 2 APP2
MAIN_D0..MAIN_D15
MAIN_A6..MAIN_A26
MAIN_D[15..0]
MAIN_A[26..6]
DARLINGTON SINK DRIVER
SIMMS 1 y 2
RESET5
2GLL2IOU0
I3
I4
XI1
XI2
XI3
XI4
1DSD_I4
1DSD_I3
1DSD_I2
1DSD_I1
2DSD_I4
2DSD_I3
2GLL2IOU0
RESET5
GLL2CCU5

P_C3 P_C2 P_C1 P_C0
R64 472
R80 103
R78 472
R79 102
R71 103
SU2IO0
BA
VCC

EEPROM*
SCLK CS DI DO
GLL2CCU5
GLL2CCU[5..0]
AT93C46-10SC
CS SCLK DI ORG NC DO
E13
EEPROMs
64/128/256 words of 16 bits each.
According to programming
3 WIRE SERIAL

RN1 303
A14
CLL2_0
CLL2_1
CLL2_2
CLL2_3
PHOTOCOUPLER
R2
R_Network08
A13
NR1 103Ω
P_C0
P_C2
P_C1
P_C3
1. Anode
2. Cathode
15. Emitter
16. Collector

CLK_GLL1
R105 330
I/OD
BIC_AOD
R106 472
CS_GLL1
WE_GLL1
RE_GLL1
SDP
MR_GLL1
ATN
ACK
REQ
Am33C93A
BUS INTERFACE CONTROLLER
(SI) I/O CLK (SCLK) A0 ALE Address Latch Enable
MSG C/D BSY SEL DRQ DACK CS WE RE SDP MR ATN ACK REQ
D0..D7 SD0..SD7 DP INTRQ (HALT)
MAIN_D[7..0]
Processor Data Bus
SD[7..0]
Small Computer System Interface Data Bus
R108 301
R107 221

RN24 RN25 RN22 RN23 RN26 RN27 RN28 RN29
221 331
I/O REQ ACK ATN
SEL BSY C/D MSG
SD3 SD4 SD5 SD6
SD2 SD1 SD0 SDP

4CH HIGH-CURRENT DARLINGTON SINK DRIVER
TD62064 C12
TD62064 C11
1DSD_I1 1DSD_I2 1DSD_I3 1DSD_I4
2DSD_I3 2DSD_I4
+12V
FB13 FB14 FB
FB18 FB16 FB17 FB15 Ferrite Bead
VSS

Capcom Kick Button Harness Connector
CN1
Conn_01x34_Male
+12V
2P_BTN6 K_H10 K_H11 K_H12 K_H13 K_H14 1P_BTN6 K_H16 1P_BTN5 K_H18 KH_2P_BTN4 K_H20 KH_1P_BTN4 K_H22 2P_BTN5 K_H24 CLL2_3 CLL2_2 CLL2_1 CLL2_0 K_H29 K_H30

FB7 FB8 FB9 FB10 F_Small
CP124 10µF
CP125 47µF

14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /iou/
File: iou.kicad_sch
Title: CP SYSTEM III
Size: A3 | Date: 14/10/22 | **Rev: José Tejada G.**
KiCad E.D.A. 10.0.4 | Id: 3/23

PWR_FLAG

Philips TDA1306T Noise Shaping Filter DAC (SOIC24).
The clock (on pin 12) measures
14.3181667MHz (42.9545/3)

14.3181667MHz

VCC

R43
101

BE

R42
472

Q8
ZR

VD3
VD-BAV99

VSS

CU383-69K
Bus-exchange switch
10 bit

BCK
SNDD
APP0
APP2
WS
SYSCLK
APP1
VSS
BE

D6

RN3
620

RN2
620

TDA1306T
Noise shaping filter
Hard-MUTE

CLKS2
CLKS1
BCK
WS
DATA
SYSCLK
APPL
APP0
APP1
APP2
APP3
TEST1
TEST2
VDDA
VDDD
VDDO
VSSA
VSSD
VSSO
FILTCR
VOR
FILTCL
VOL
VREF

C5
DAC

CP153
22µF
CP154
22µF
CP155
22µF

C143
102J

C144
102J

CP123
10µF

R38
7.3kΩ
R37
4.7kΩ
R35
4.7kΩ
R34
7.3kΩ

LM833N
A4B
Opamp_Dual

CP131
22µF
CP135
22µF
CP132
22µF

R_POT_5PINS
RV1

CP134
22µF

Opamp_Dual
A4A

R20
1.8kΩ
R19
1kΩ
R3
100Ω
CP138
22µF
CP133
22µF
0.003Ω
EXC2
R_Shunt_US
J2
AudioJack2
RS2
100kΩ

R27
9kΩ

CP137
22µF

A5B
LM833N
Opamp_Dual

R17
2.3kΩ
R18
2.3kΩ

CP143
3.3µF

TA8201AK B3
IN
NF
R.F
OUT1
OUT2

R4
5.6kΩ
CP139
33µF
CP140
220µF

R9
0.3Ω
R8
0.3Ω
CP145
154J
CP146
154J

FB6
F_Small
FB5
F_Small
SPEAKER(+)
SPEAKER(-)
LS1
Speaker

HS1
Heatsink_Pad_2Pin

Opamp_Dual
A5A

R22
1.8kΩ
R21
1kΩ
CP130
22µF

R11
100Ω
CP136
22µF
0.003Ω
EXC1
R_Shunt_US
R1
100kΩ
J1
AudioJack2
Optional RCA Left/Right Audio Out Connectors

R28
9kΩ

+12V
FB1
F_Small
R12
24Ω

+12V
FB2
F_Small

QB6
Q_NPN_CBE
CP148
22µF

QA6
Q_NPN_CBE
CP149
22µF
CP150
22µF

CP147
22µF

14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /sound/
File: sound.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 4/23

G2
DL-2729_PPU
Picture Processing Unit
SD10-505
Main graphics chip.

Synchronization Supply Unit*?

Buffer Frame Memory Data Bus

There is no certainty that it is Data Bus

There is no certainty that it is Address Bus

Buffer Frame Memory Address Bus

Color Palette

Bus Exchange Switch

Row address strobe

SPRITE ADDRESS BUS
*Buffer Line Driver

SCROLL ADDRESS BUS

Output enable

Write enable

Upper column address strobe

Lower column address strobe

system clock 256fs or 384fs

60.000MHz

14.3181667MHz

42.9545MHz/3

7100A OSC 9640

CU383-69K

MB814260-70PJ

FAST PAGE MODE DYNAMIC RAM CMOS 256K X 16 BIT

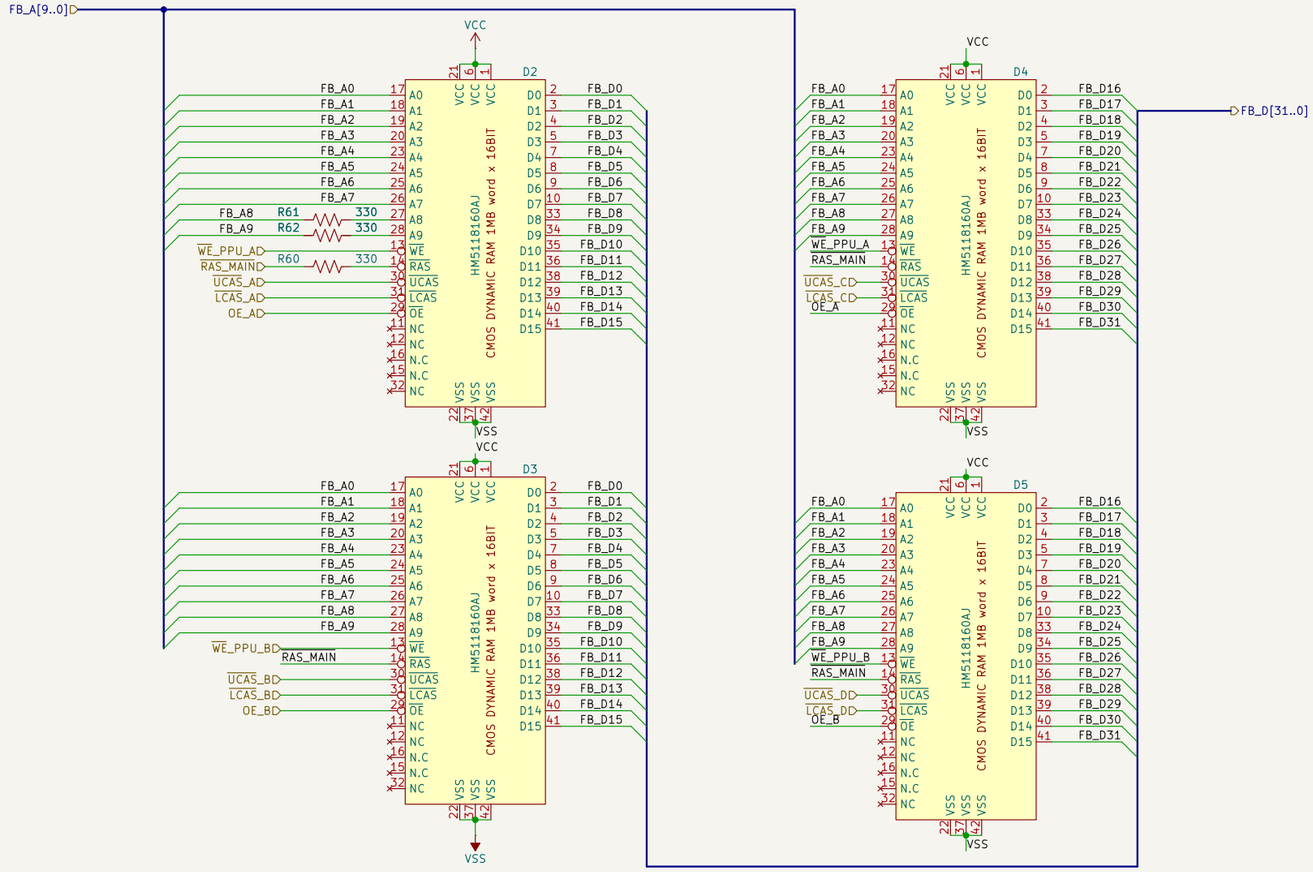
Bus-exchange switch 10BIT

File: memories_fb.kicad_sch

File: buffer_line_driver.kicad_sch

14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /ppu/
File: ppu.kicad_sch
Title: CP SYSTEM III
Size: A3 | Date: 14/10/22 | **Rev: José Tejada G.**
KiCad E.D.A. 10.0.4 | Id: 5/23

MEMORIES FRAME BUFFER

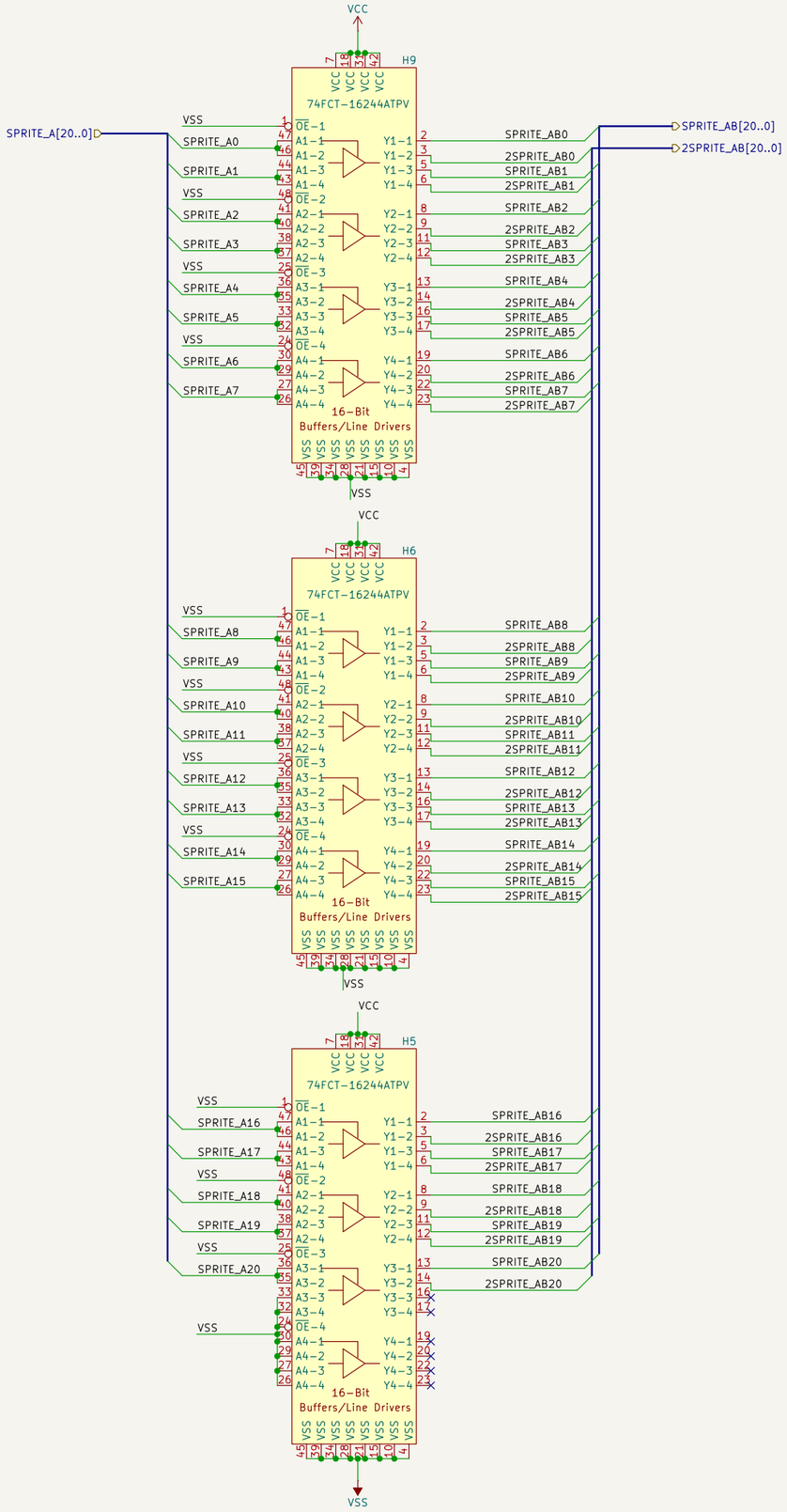


14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /ppu/memories_fb/
File: memories_fb.kicad_sch

Title: CP SYSTEM III

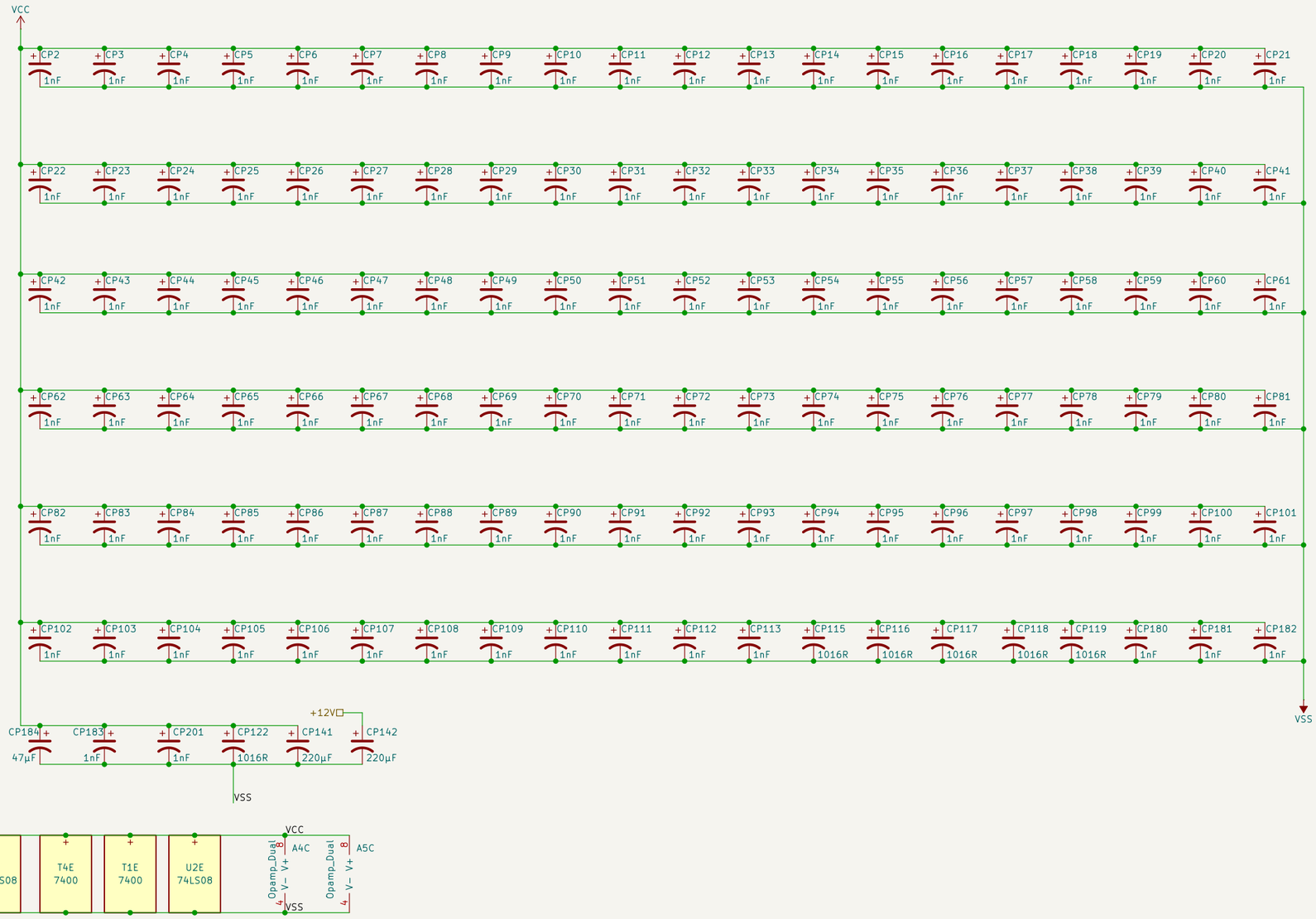
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 20/23



14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /ppu/buffer_line_driver/
File: buffer_line_driver.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 21/23



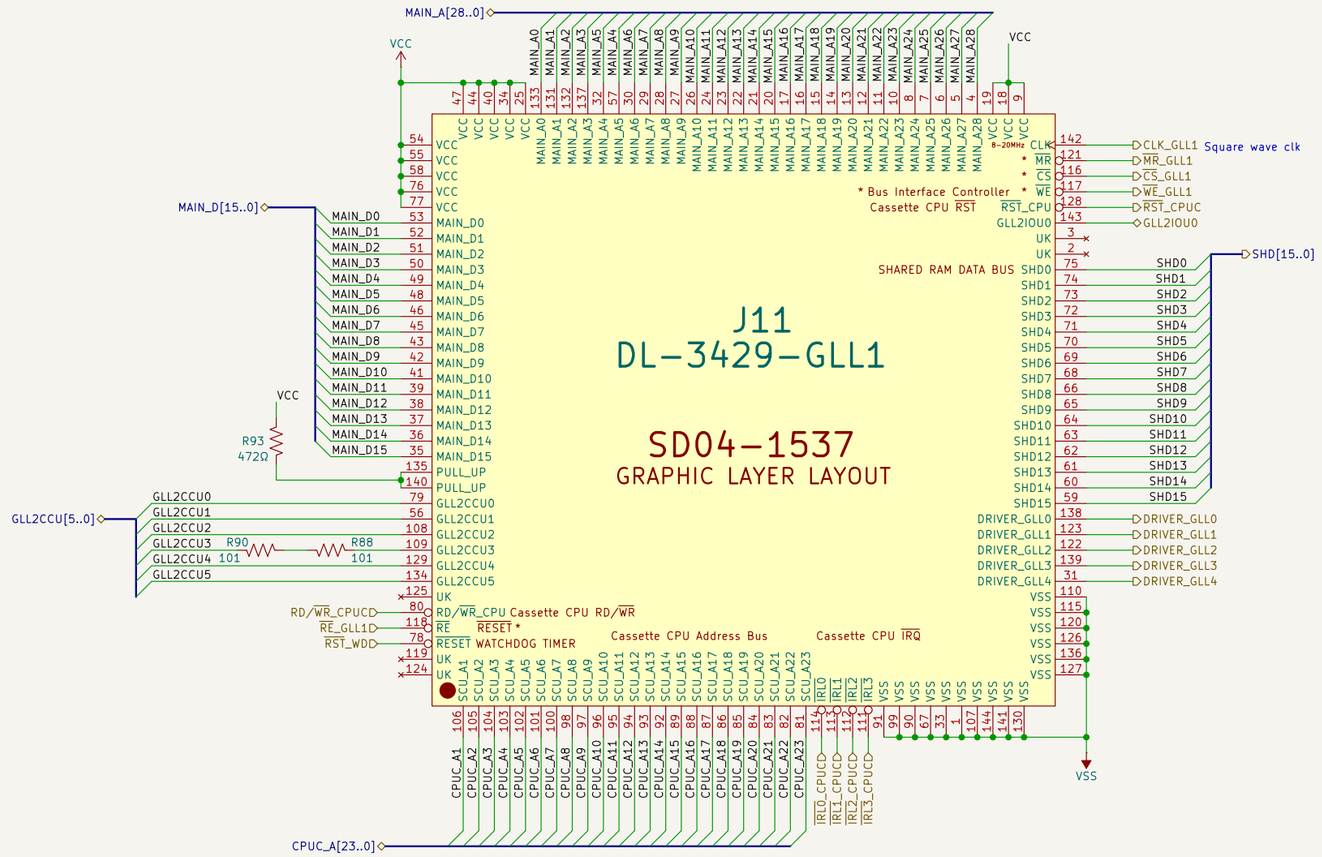
14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /capacitors_uu/
File: capacitors_uu.kicad_sch

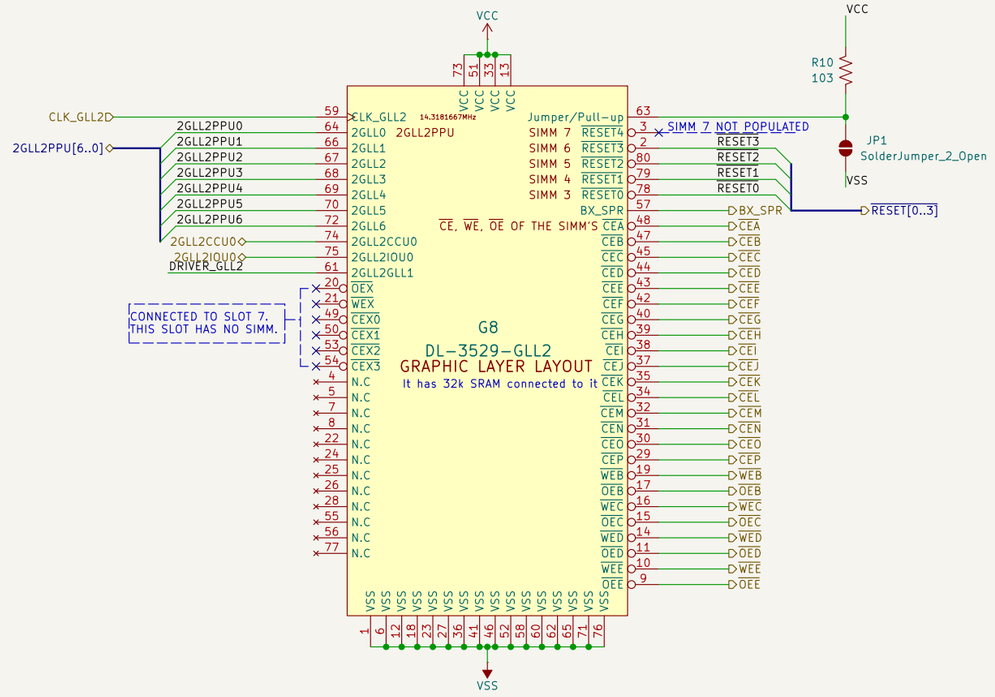
Title: CP SYSTEM III

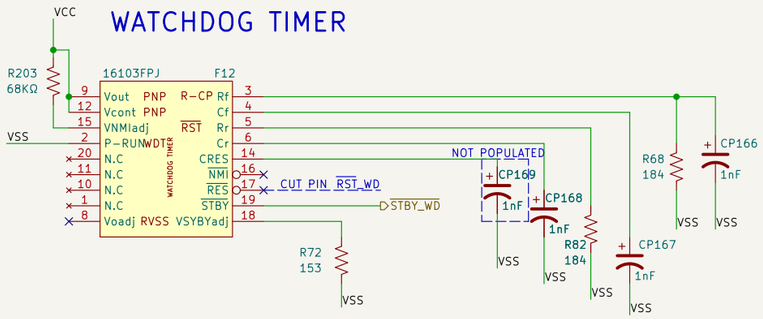
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 6/23

GRAPHIC LAYER LAYOUT 1

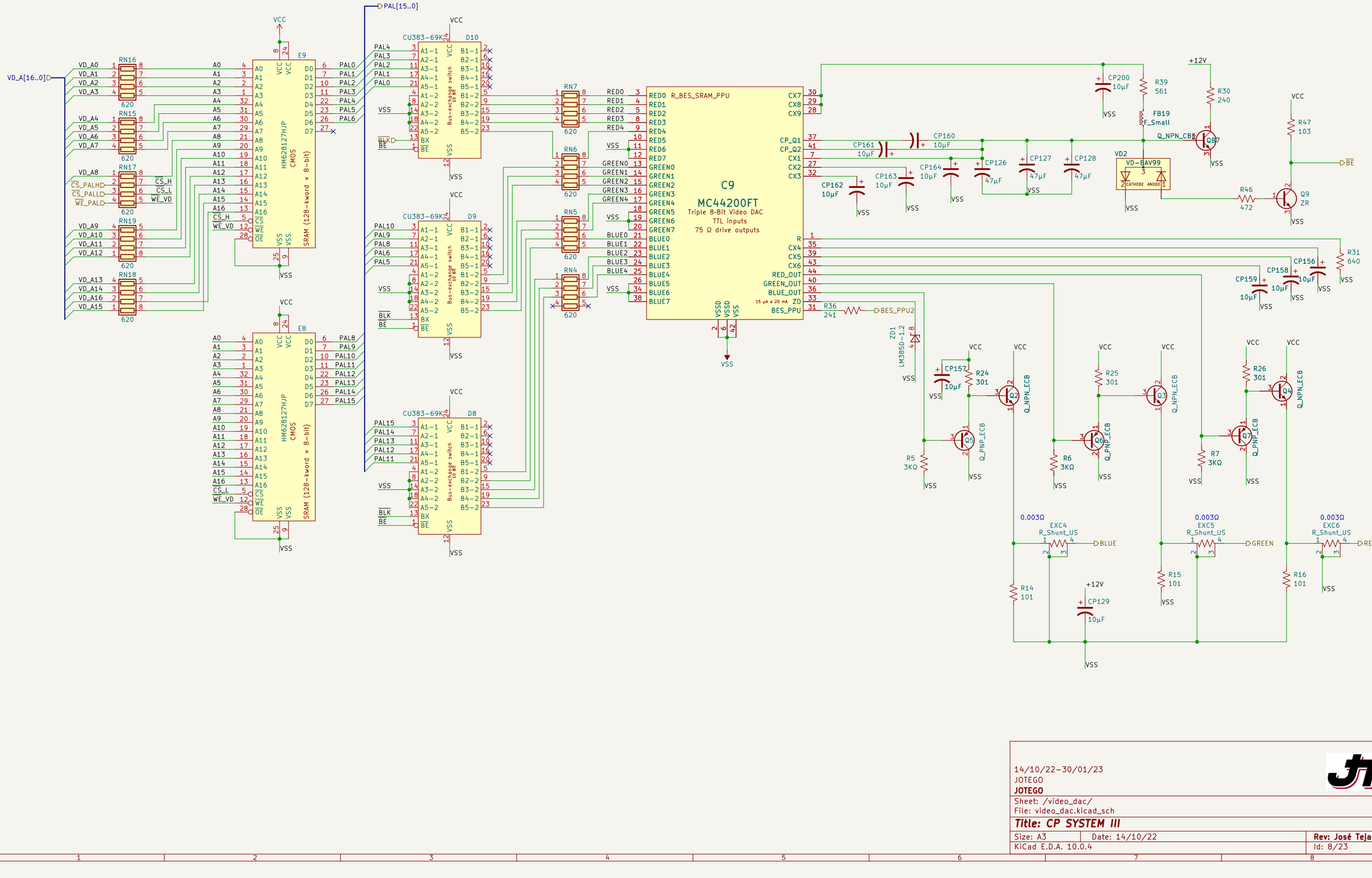


GRAPHIC LAYER LAYOUT 2





14/10/22-30/01/23		
JOTEGO		
JOTEGO		
Sheet: /objects/		
File: objects.kicad_sch		
Title: CP SYSTEM III		
Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 7/23



14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /video_dac/
File: video_dac.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 8/23

G6
DL-3329-SSU
SD04-1536
SYNCHRONIZATION SUPPLY UNIT?

Clocked at 21.47725MHz (42.9545/2)

MAIN_D[14..0]
Possibly this is the address bus.

MAIN_A[24..6]
Possibly this is the data bus.

SSU_A[16..0]

D[7..0]
A[14..0]

F9
N341256SJ_20
CMOS SRAM
256K-BIT

VD_A[16..0]
to video DAC

R85
472

R29
100

Q1
ZR

R13
47Ω

R23
47Ω

74LS08
U1A

14/10/22-30/01/23
JOTEGO
JOTEGO

Sheet: /ssu/
File: ssu.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 9/23

J13
DL-2829-CCU
COMMUNICATION CONTROL UNIT?

Cassette CPU RD
Cassette CPU CS
Cassette CPU Address Bus
*ENABLING SIMMS MEMORIES - S1 Y 2
*SOUND RAM SIMMS ADDRESS BUS
CASSETTE RAM CE
* VOLTAGE MONITOR
* SOUND RAM SIMMS DATA BUS
SHARED RAM DATA BUS
Cassette CPU Data Bus
ONLY CONNECTED TO PIN52 OF THE FEMALE SLOT

Shared RAM

MB814260-70PJ
FAST PAGE MODE DYNAMIC RAM CMOS 256K X 16 BIT

VMON_XXN
VOLTAGE DETECTOR
INCIERTO

14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /ccu/
File: ccu.kicad_sch
Title: CP SYSTEM III
Size: A3 | Date: 14/10/22 | **Rev: José Tejada G.**
KiCad E.D.A. 10.0.4 | Id: 10/23

No device mounted
there is only the footprint

CP189 X
CP187 1µF
CP185 47µF
CP186 1µF
CP205 101
CP190 X
CP188 1µF
CP176 1µF
CP194 1µF
CP177 X
CP175 1µF
CP195 1µF
CP196 1µF
CP174 1µF
CP191 1µF
CP192 1µF

RR6 103
RR3 103
RR19 X
RR24 103
RR23 103
RR22 103
RR21 103
RR20 103
RR2 105
RR16 302
RR17 302
RR11 472
RR12 472
RR13 472
RR14 472
RR15 472
RR9 472
RR10 472
RR18 301
RR8 102
RR7 104
RR5 473
RR4 103
CP178 1µF
CP206 1µF
CP193 1µF

BT1 Battery
STANDBY
V.D2
V.D1
BD53E42
TRIP POINT 4,2V
RST_WD
RST_CPUC
NOT POPULATED
Capacitor connection terminal for output delay time
DA1 B3 61t
Small Signal Zener Diode?
regulador de voltaje?
pending review here
pending review here
pending review here
pending review here
REDUCED INSTRUCTION SET COMPUTER
VCC
VSS
+5V
VCC(PLL)
CPUC_D[31..0]
CPUC_A[26..0]
CKIO_CPUC
X1 Crystal 6.25SC6L
CASSETTE MALE CONNECTOR
CNN5
Conn_02x60_Male
HD6417604F28_SH2
SCU3
CPU
(SCU) System Control Unit?
SLAVE?
Phase Locked Loop(PLL)
*External interrupt source input
***Nonmaskable interrupt request
**Interrupt vector fetch cycle
Clock pause request input CKPREQ/CKM
Clock pause acknowledge output CKPACK
Operating mode pin
External capacitance pin for PLL
Serial clock input/output
Serial data output
Serial data input
Free-running timer clock input
Free-running timer input
Free-running timer output
Watchdog timer output WDTOVF
Bus request in slave mode, bus grant in master mode BREQ/BGR
Bus acknowledge in slave mode, bus request in master mode BACK/BRLS
Hardware wait request WAIT
Synchronous DRAM clock enable control CKE
Read pulse RD
Least significant byte selection signal for memory CASLL/DQMLL/WE0
Third byte selection signal for memory CASHH/DQMUU/WE1
Second byte selection signal for memory CASHH/DQMUU/WE2
Most significant byte selection signal for memory CASHH/DQMUU/WE3
CAS for synchronous DRAM, OE for pseudo-SRAM CAS/OE
RAS for DRAM and synchronous DRAM, CE for pseudo-SRAM RAS/CE
Read/write RD/WR
Chip select CS3 CS2 CS1
* DMAC request
** DMAC acknowledge
RD_CPUC
CPUC_WE[3..0]
RD/WR_CPUC
BS_CPUC
CPUC_CS[3..0]
SCK_CPUC
RxD_CPUC
FTCI_CPUC
FTI_CPUC
WAIT_CPUC
CE_FRAM
MD
BREAKDOWN?
QQ2 AEY
QQ1 BR39
memories_scu
CER_CSTD
WE_CMOS
RST_CMOS
BYTE
OE_CMOS
CPUC_D[31..0]

14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /cassette/
File: cassette.kicad_sch
Title: CP SYSTEM III
Size: A3 | Date: 14/10/22 | **Rev: José Tejada G.**
KiCad E.D.A. 10.0.4 | Id: 11/23

NOT POPULATED

This device does not support the 10% VPP option at extended temperatures, as it is in the commercial data sheet.
There is a table that specifies a tolerance of 5% VPP only.

CPUC_A[26..0]

CPUC_D[31..0]

Z5 — SMART 3 ADVANCED BOOT BLOCK — 28F400-200BX/BL — FLASH MEMORY FAMILY
4MB (256Kb x 16bits) 8MB (512Kb x 16bits) 16MB (1024Kb x 16bits)
Vpp Peak to Peak Voltage
DI/O_A_MUX D15/A-1
BYTE Selects 8-bit or 16-bit Mode
5 Volt Sector Erase

CP179 10µF

Z2 — 29F400TA-90PF — CMOS FLASH MEMORY — 4MB (512Kb x 8bits)
DI/O_A_MUX D15/A-1
READY/BUSY STATUS _O RY/BY
BYTE Selects 8-bit or 16-bit Mode
5 Volt Sector Erase

CCP180 1µF

7400 T1A — WAIT_CPUC

7400 T1D — CPUC_CS1D

7400 T1C — CE_FRAM

CONNECTED IN FEMALE SLOT:
WE_CMOS TO PULL UP
RST_CMOS TO VCC
OE_CMOS TO VSS
BYTE TO VSS

FM1208S-250SC Z6 — 4096-BIT NONVOLATILE FERROELECTRIC RAM
Mode Control Input

14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /cassette/memories_scu/
File: memories_scu.kicad_sch

Title: CP SYSTEM III

Size: A3	Date: 14/10/22	**Rev: José Tejada G.**
KiCad E.D.A. 10.0.4		Id: 22/23

HARDWARE DESCRIPTION

Main CPU : Main CPU: Hitachi HD6417099 (SH-2) at 25 MHz
Storage : SCSI CD Rom, RAM and Flash ROM.
RAM : SIMMs on the left will physically hold 16M of data each (each has 8x 16M Flash ROMs, there's 4 SIMMs plugged in on this board). The SIMMs on the right will physically hold 8M each (there's 4 Flash ROMs on each board and there are 2 SIMMs plugged in)
OSC : 60MHz. 42.9545MHz, XTAL :3.6864MHz

**Sound chip: 16-channel 8-bit sample player, stereo*
**Maximum color palette: 16 million shades[5]*
**Maximum number of colors on screen: 32,768[6] (15-bit colour, 555 RGB)*
**Palette size: 131,072 pens*
**Colors per tile (backgrounds / sprites): 64 (6 bits per pixel) or 256 (8 bits per pixel), selectable*
**Colors per tile (text overlay): 16 (4 bits per pixel)*
**Maximum number of objects: 1024, with hardware scaling[6]*
**Scroll faces: 4 regular + 1 text overlay 'score screen' layer*
**Scroll features: Horizontal & vertical scrolling, linescroll, linezoom[6]*
**Framebuffer zooming*
**Color blending effects*
**Hardware RLE decompression of 6 bpp and 8 bpp graphics through DMA*
**Resolution, pixels: 384×224 (standard mode) / 496×224 (widescreen mode)*

KNOWN CP SYSTEM III (CPS3) GAME LIST

All six games are developed by Capcom and are all head-to-head fighting games in a list of games that appears to be as small as the CP System Dash library.

English title	Release date	Japanese title
Red Earth	1996-11-21	*War-Zard* (ウォーザード)
Street Fighter III: New Generation	1997-02-04	*Street Fighter III* (ストリートファイターIII)
Street Fighter III 2nd Impact: Giant Attack	1997-09-30	*Street Fighter III 2nd Impact* (ストリートファイターIII 2nd Impact)
JoJo's Bizarre Adventure	1998-12-02	*JoJo no Kimyō na Bōken* (ジョジョの奇妙な冒険)
Street Fighter III 3rd Strike: Fight for the Future	1999-05-12	*Street Fighter III 3rd Strike* (ストリートファイターIII 3rd Strike)
JoJo's Bizarre Adventure: Heritage for the Future	1999-09-13	*JoJo no Kimyō na Bōken Mirai e no Isan* (ジョジョの奇妙な冒険 未来への遺産)

Font:
The Museum Arcade System 16

14/10/22-30/01/23
JOTEGO
JOTEGO
Sheet: /h_description/
File: h_description.kicad_sch
Title: CP SYSTEM III
Size: A3 | Date: 14/10/22 | **Rev: José Tejada G.**
KiCad E.D.A. 10.0.4 | Id: 23/23